CT scan reconstruction using machine learning

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abstract

This will be the abstract

Introduction

This will be the introduction

Theory

Basics CT scans

A CT scan is an imaging technique which uses x-rays sto create cross-sectional images of the body. A CT scan machine consists of a source and a detector placed at a distance from eachother. During the imaging procedure, both the source and the detector will rotate around the object to be imaged. The source will emit x-rays which will partially travel through the object towards the detector. The detector will measure the spectrum of transmitted photons on the other side of the object to be imaged. The spectrum is shaped by how much of the x-rays are being absorbed or scattered by the body, which can be described as the number of detected photons $\hat{y}_i$, generally given as

$$\hat{y}_i = N_0 \cdot \exp\left(-\int \mu(x, E)\, dx\right) \tag{1}$$

where N_0 is the number of the x-rays entering the body, the subscript i denotes the location along the detector and the integral in the exponent represents the cumulative attenuation $\mu(x, E)$ of the x-ray beam along it's path x, where the linear attenuation coefficient $\mu(x, E)$ changes throughout the body (Kamalian, Lev, and Gupta, 2016). When measureing $\hat{y}_i$ for a set of angles around the object to be imaged, a sinogram can be constructed by plotting $\hat{y}_i$ aginst the angle.

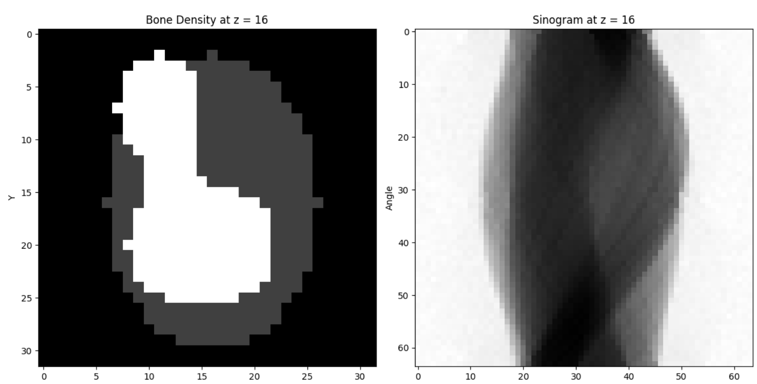


Figure 1. *left:* a 2D slice of the phantom the colour gradient representing bone density. *right* corresponding sinogram with 64 projection angles and a detector size in the y-direction of 64 pixels - say what energy

Both a phantom and a corresponding sinogram can be seen in Figure 1. Here the phantom shows the simulated bone density of an object, with lighter coloured areas having a higher bone density. On the right is the corresponding sinogram which consists of 64 measurements of $\hat{y}_i$ at different angles.

Cone beam CT scan

The sinogram in figure Figure 1 is true for a 1D detector, where we only measure along the Z-axis of the phantom. A cone beam CT scan uses a source which emits in a cone like shape and a 2D detector to measure the projections.

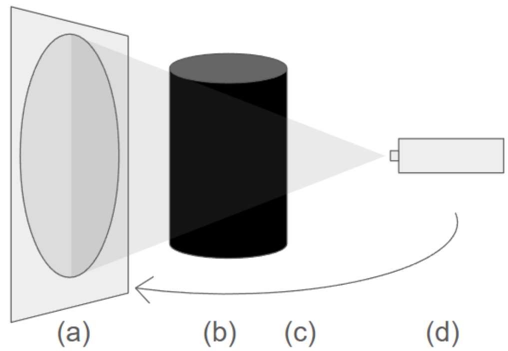


Figure 2. A schematic drawing of a cone beam CT scan procedure with (a) the detector measuring the projections in 2D, (b) the phantom, (c) the direction of rotation for both detector and the source and (d) the x-ray source.

For data reconstruction, a more complex backprojection algorithm needs to be used as we introduce the angle between the incomming x-rays onto the detector and the detector itself as a variable.

Detector types

Modern CT scan machines currently use an energy integrating detector (EID) (Marth et al., 2023). An EID measures the intensity of the incomming x-rays by a two-step process: scintillation followed by photodetection.

There are 2 main issues with these detectors we will focus on:

- The contrast between different soft tissues is often insufficient
- images are not tissue-type specific

These detectors measure by the energy-integrated signal of photons (Taguchi and Iwanczyk, 2013) which loses all energy dependent information.

To be able to simulate a body in a CT scan, material decompositon (Mechlem et al., 2018) can be used. The linear attenuation coëfficient at location x will be

Material decomposition

$$\mu(E) = \sum_{b=1}^{B} A_b f_b(E) \tag{2}$$

where A_b is the line integral of material b and $f^b(E)$ describes the attenuation coefficient of the basis material b. Combining the equation (**??**) and (2) gives an expression for the photon count at the detector:

something relating I to y_i

$$\hat{y}_i = \int_0^{\infty} \phi_{\text{eff},i}(E) \exp\left(-\sum_{b=1}^{B} A_b^i f_b(E)\right) dE \tag{3}$$

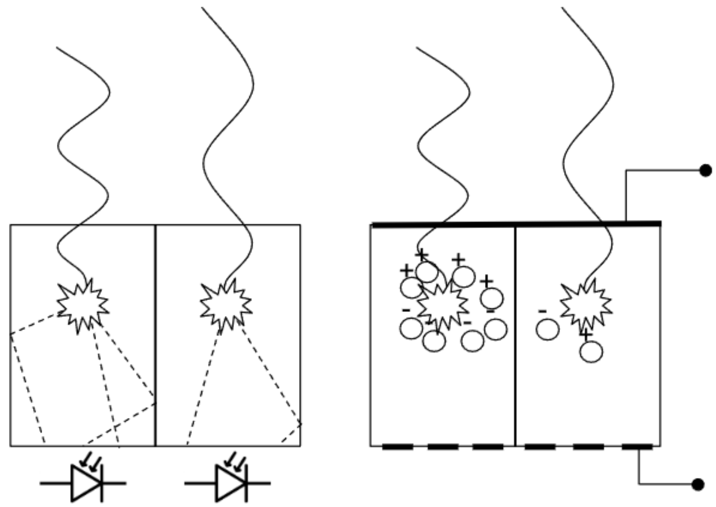
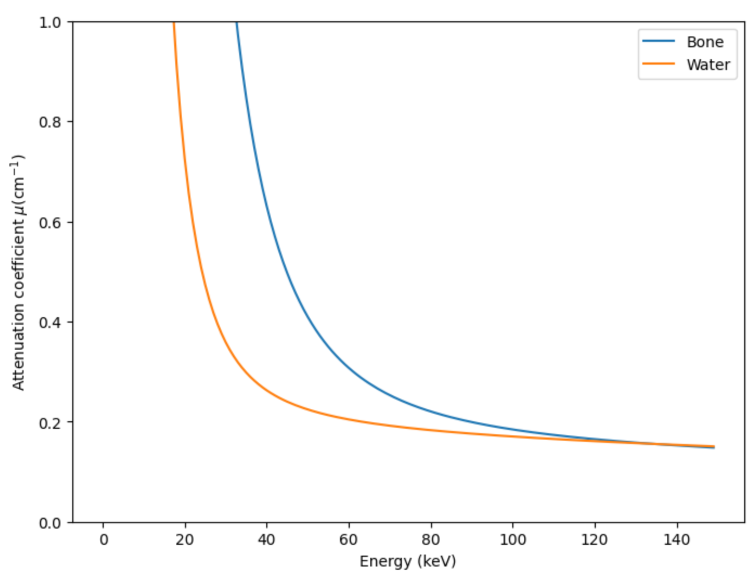


Figure 3. *left:* energy integrating detector. *right:* photon counting detector



The attenuation coefficient of bone (blue) and water (orange) plotted against the energy of the incomming photons.

During a CT-scan $\hat{y}_i$ is measured for a set of angles. Each measurement results in a 1D array for a 2D slice.

Recent breakthroughs in photon counting detectors make use of equation (2), where the linear attentuation coefficient depends on the energy level. The photon counting detectors (pcd) allow the classification of measured photons into energy bins (Taguchi, Ballabriga, et al., 2022).

Projection algorithms

- brief overview backprojection (Zeng, 2001)
- list issues
- brief overview itterative approach (with references to Mechlem)
- positives
- negatives

part of abstract: "g. They require a specific reconstruction process, consisting of two steps: material decomposition and tomographic reconstruction. Image-based methods perform reconstruction, then decomposition, while projection-based methods perform decomposition first, and then reconstruction. As an alternative, 'one-step inversion' methods have been proposed, which perform decomposition and reconstruction simultaneously. Unfortunately, one-step methods are typically slower than their two-step counterparts, and in most CT applications, reconstruction time is critical. This paper therefore proposes to compare the convergence speeds of five one-step algorithms." (Mory et al., 2018)

itterative unet (Ge et al., 2023)

machine learning - neural networks

Neural networks are a subset of machine learning and play a crucial role in deep learning algorithms. In the basis, neural networks consist of a number of nodes. A set of nodes are combined into layers where all layers are connected by weights. Starting from the input layer, each input node corresponding with a number is connected to every node in the first layer with a weight. to calculate the value of a node in the next layer, all node from the previous layer are multiplied by the weight connecting them. Stacking layers after eachother creates a neural network.

Neural networks learn by backpropagation. Backpropagation looks at the error between the output of the network and the desired output. This error is generally backpropogated through the network by looking at how each weight needs to change to minimise the error.

machine learning - CNN

A convolutional neural network can be used to more accuratly extract features from images for processing. A convolutional neural network consist of a kernel and an amount of channels. The kernel has an equal number of dimensions as the input data however smaller which consists of weights. The kernel will move across the input data performing element wise multiplication with the part of the input they cover, extracting information like edges, textures and shapes (Yamashita et al., 2018).

- UNet github(Ronneberger, Fischer, and Brox, 2015)

Machine learning - Attention

An attention mechanism is a technique used in machine learning which is used to allow machine learning models to attend to the most relevant parts of the input data (Vaswani et al., 2017).

- attention map?
- How we want to map the sinogram to the image, which is non-local (Vaswani et al., 2017)

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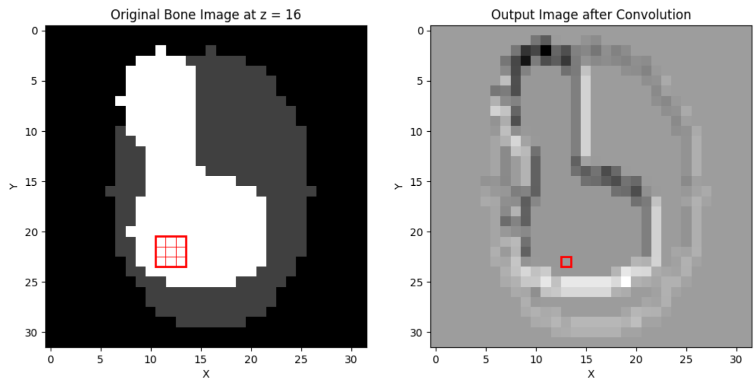
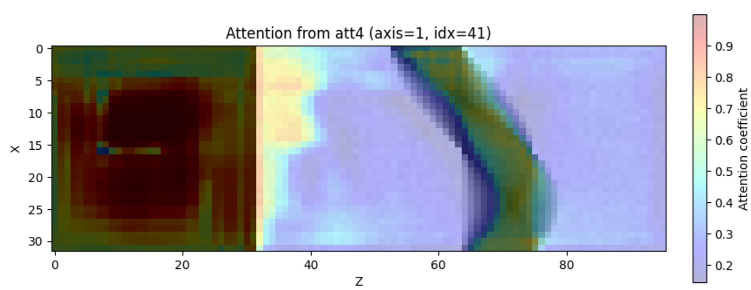


left : original phantom with a 2D 3x3 randomly initialised kernel drawn on top. each individual square represents a pixel, with the outer square representing the kernel. *right :* phantom after 2D convolution layer has been applied.



: label : attention_network An attention map showing how an attention mechanism can be used to have the model focus more on important features, which in this case is the sinogram.

Machine learning - UNet with attention

How multiple convolutions and attention networks can be combine to edit images

Method

acquiring data

A machine learning model will be trained to convert an image from sinogram space to image space. This will be done according to the steps taken in an iterative approach (Mechlem et al., 2018) and the model will be trained on the intermediate steps calculated. To be able to train the model we need 2 sorts of data:

- projection matrix
- images during reconstruction The projection matrix will have the dimensions of the number of projections x y pixels detector x z pixels detector. The images during reconstruction refer to the intermediate images between each iteration of the iterative algorithm. As both parts of the data have a 3D format, the images will be padded with 0's to match the y-dimension of the projection matrix. The image and the projection are appended in the z-direction.

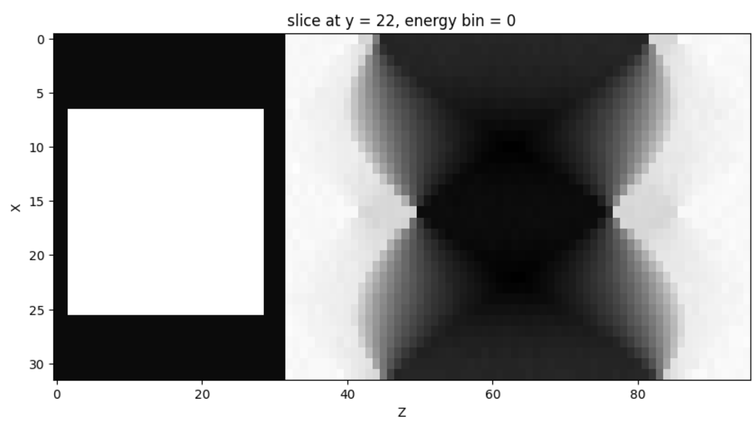


Figure 4. *left:* example of input data, where on the left we see a square which is a cross section of a 3D cylinder. On the right corresponding sinogram. Axes don't line up!

- how many mechlem runs
- how many iterations
- stride of the input images

Attention based UNet

UNet has already been used to segmented, based on github

what network has finally been constructed etc

curriculum learning

Collected data {number from notebook}

Training on phantoms

Collected {number from notebook} sets of training data which will be implemented with the following parameters:

results

plot about speedup (time vs pixel size)

either 2D or 3D

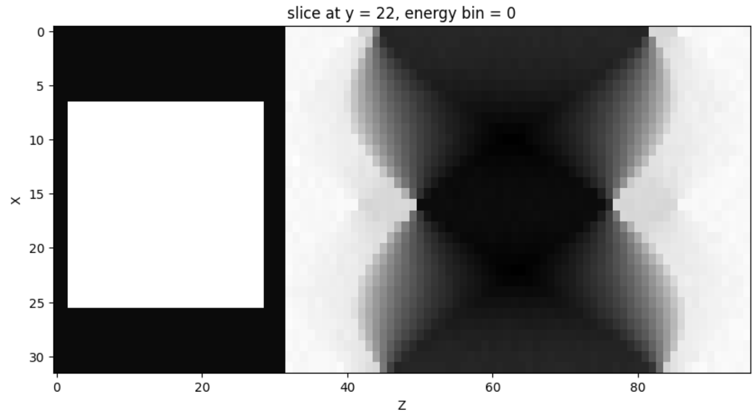


Figure 5. *left:* example of input data, where on the left we see a square which is a cross section of a 3D cylinder. On the right corresponding sinogram. Axes don't line up!

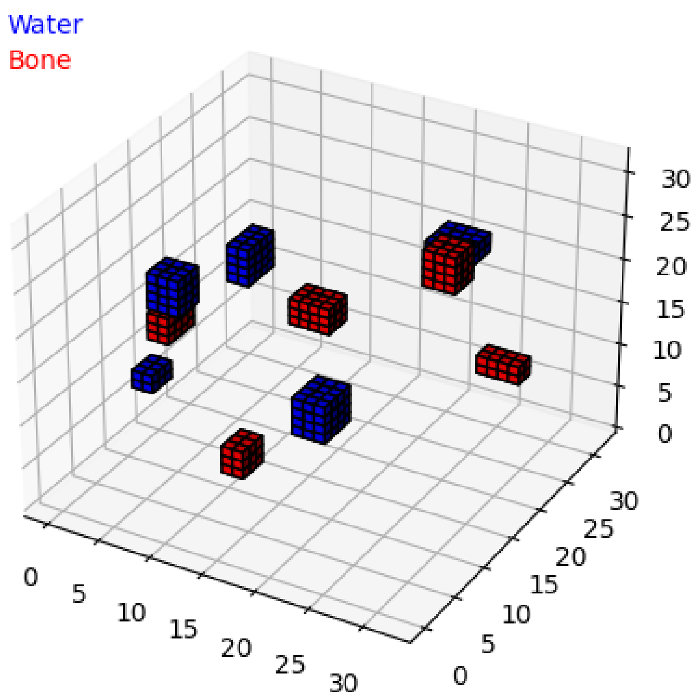


Figure 6. *left:* Single data set used for curriculum learning.

plot about results for 32x32 simple phantom

plot harder phantom (3 features or more)

Discussion

non-machine learning speedup

pytorch is highly optimized, mechlem is not

pixel size and number of projections necessary

There will not be a linear relationship between the amount of projections necessary and the amount of pixels in the image. will this have an impact?

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